

Scalability & Future Plan

Date	14 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines realistic scalability considerations and future enhancements for the Personalized Networking Assistant based on its current React, TypeScript, Node.js, Express, SQLite, Google Gemini, Wikipedia, and Render architecture.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	SQLite is suitable for the current project scale but is not ideal for high-concurrency multi-instance deployment.	Can limit horizontal scaling and concurrent write capacity as the user base grows.	High
2	AI generation and factual lookup depend on external Gemini and Wikipedia services.	Network latency, provider limits, or temporary service issues can affect response time and availability.	High
3	The current application does not include direct integrations with professional networking or event platforms.	Users must manually enter profile and event context instead of importing it from external services.	Medium

Scalability Plan:

S.No	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js/Express application deployment.	Scale stateless API instances behind a load balancer and use managed hosting infrastructure when traffic grows.
2	Data Storage	SQLite database with optional persistent disk on Render.	Migrate to managed PostgreSQL for stronger concurrency, backups, and multi-instance deployment.
3	Performance	Live Gemini generation and on-demand Wikipedia lookup.	Add safe caching, request optimization, background processing where appropriate, and provider-aware rate handling.
4	Security	Server-side secrets, bcryptjs password hashing, and parameterized SQL operations.	Add production HTTPS enforcement, stronger session management, rate limiting, monitoring, and role-based controls if required.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve production reliability with managed PostgreSQL migration, stronger monitoring, rate limiting, and refined deployment configuration.	Future Enhancement	Supports larger user loads, better concurrency, operational visibility, and more reliable persistence.
Phase 2	Add optional integrations with event, calendar, or professional profile services where supported and authorized.	Future Enhancement	Reduces manual context entry and enables richer event-aware personalization.
Phase 3	Expand the assistant with multilingual generation, additional trusted factual sources, richer analytics, and optional voice or mobile-friendly interaction.	Future Enhancement	Improves accessibility, factual coverage, usability, and support for a broader range of networking scenarios.